

UMC 0.11um LOGIC/MIXED_MODE AI Advanced Enhancement MOM Capacitors SPICE Model Document

**Version 0.2 Phase 2
2020/05/26**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
T.ApB	2007/2/2	Hey Jong Su	Original release of Paper model(without Si data)
0.1_p1	2010/03/04	Chih hung Chuang	1. Copy from MM130E_MOM_V041 model and shrink to 90% geometry size. 2. Follow New DSM version definition from 2009/11/30.
0.2_p1	2010/7/27	Eddie Chang	Add stripe MOM structure
0.2_p2	2020/5/26	Min Chi Huang	Modify document typo 1. Symmetric: minimum lf from 4.4u to 2.4u and minimum nf of from 14 to 8. 2. Asymmetric: minimum lf from 4.4u to 2.4u and minimum nf of from 15 to 9. 3. Stripe: minimum lf from 4.4u to 2.52u and minimum nf of from 14 to 8.

1.2 General Information

This document serves as a reference to the design works of products that will be manufactured by UMC using UMC's L110E Mix-Mode/RF Process.

1.3 Model Usage

The compact model is provided in various formats. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.09

Cadence SPECTRE version 6.2.1.299

Mentor Graphics ELDO version 6.11_1.1 (2008.1)

*Note: Known difference among different simulators:

Since HSPICE and ELDO don't directly support voltage coefficients in capacitor models, there is no voltage coefficients included in HSPICE and ELDO models.

Spectre model has included voltage coefficients for MOM capacitors.

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

TT: Typical Capacitance value

SS: Maximum Capacitance value

FF: Minimum Capacitance value

Model Parameters explanations

c : main capacitance

nf : number of fingers

nm: number of main capacitance layers

m : mesh numbers

bm : bottom layer

shielding_flag : 1 for with shielding layer structure

0 for without shielding layer structure

metal_ring_flag : 1 for with metal ring structure

0 for without metal ring structure

presim_flag : 1 for pre-layout simulation

0 for post-layout simulation

The instant parameters for MOM capacitance simulation could be c, nf, nm, bm or l (before-shrink), nf, nm, bm. It is meaningless if instant parameters c and l are used at the same time.

Follow these steps to invoke the model file during simulation.

1. For Cadence Spectre

-Copy L110AE_MOM_V021.lib.scs & L110AE_MOM_V021.mdl.scs into your simulation directory.

-Add the following statement in the input file to the simulator for 90% dimension shrinkage.
SetOption 1 options scale=0.9

include "/L110AE_MOM_V021.lib.scs" section=CornerSection

, where CornerSection could be tt, ss, ff.

-Follow the example below to call the model

For non-mesh symmetry MOM capacitors

Non-shielding layer

x1 (port1 port2 well) momcaps_sy_mml110ae c=100f nf=20 nm=5 m=4 bm=1

shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0

or

x1 (port1 port2 well) momcaps_sy_mml110ae l=3u nf=20 nm=5 m=4 bm=1

shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0

With shielding layer

x1 (port1 port2 well) momcaps_sy_mml110ae c=100f nf=20 nm=5 m=4 bm=2

shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

or

x1 (port1 port2 well) momcaps_sy_mml110ae l=3u nf=20 nm=5 m=4 bm=2

shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh symmetry MOM capacitors

Non-shielding layer

x1 (port1 port2 port3 port4 well) momcaps_symesh_mml110ae c=100f mh=1 nf=20 nm=5

m=4 bm=1 shielding_flag=0 presim_flag=1

or

x1 (port1 port2 port3 port4 well) momcaps_symesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4

bm=1 shielding_flag=0 presim_flag=1

With shielding layer

x1 (port1 port2 port3 port4 well) momcaps_symesh_mml110ae c=100f mh=1 nf=20 nm=5

m=4 bm=2 shielding_flag=1 presim_flag=1

or

x1 (port1 port2 port3 port4 well) momcaps_symesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4
bm=2 shielding_flag=1 presim_flag=1

For non-mesh asymmetry MOM capacitors

x1 (port1 port2 well) momcaps_as_mml110ae c=100f nf=20 nm=5 m=4 bm=1
metal_ring_flag=0 or 1 presim_flag=1

or

x1 (port1 port2 well) momcaps_as_mml110ae l=3u nf=20 nm=5 m=4 bm=1
metal_ring_flag=0 or 1 presim_flag=1

For mesh asymmetry MOM capacitors

x1 (port1 port2 port3 port4 well) momcaps_asmesh_mml110ae c=100f mh=1 nf=20 nm=5
m=4 bm=1 presim_flag=1

or

x1 (port1 port2 port3 port4 well) momcaps_asmesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4
bm=1 presim_flag=1

For non-mesh stripe MOM capacitors

Non-shielding layer

x1 (port1 port2 well) momcaps_stripe_mml110ae c=100f nf=20 nm=5 m=4 bm=1
shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0

or

x1 (port1 port2 well) momcaps_stripe_mml110ae l=3u nf=20 nm=5 m=4 bm=1
shielding_flag=0 presim_flag=1 metal_ring_flag=1 or 0

With shielding layer

x1 (port1 port2 well) momcaps_stripe_mml110ae c=100f nf=20 nm=5 m=4 bm=2
shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

or

x1 (port1 port2 well) momcaps_stripe_mml110ae l=3u nf=20 nm=5 m=4 bm=2
shielding_flag=1 presim_flag=1 metal_ring_flag=1 or 0

For mesh stripe MOM capacitors

Non-shielding layer

x1 (port1 port2 port3 port4 well) momcaps_stripemesh_mml110ae c=100f mh=1 nf=20
nm=5 m=4 bm=1 shielding_flag=0 presim_flag=1

or

x1 (port1 port2 port3 port4 well) momcaps_stripemesh_mml110ae l=3u mh=1 nf=20 nm=5
m=4 bm=1 shielding_flag=0 presim_flag=1

With shielding layer

x1 (port1 port2 port3 port4 well) momcaps_stripemesh_mml110ae c=100f mh=1 nf=20
nm=5 m=4 bm=2 shielding_flag=1 presim_flag=1

or

x1 (port1 port2 port3 port4 well) momcaps_stripemesh_mml110ae l=3u mh=1 nf=20 nm=5
m=4 bm=2 shielding_flag=1 presim_flag=1

2. For Hspice

- Copy L110AE_MOM_V021.lib & L110AE_MOM_V021.mdl into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9
.lib "/L110AE_MOM_V021.lib" CornerSection
, where CornerSection could be tt, ss, ff.
- Follow the example below to call the model

For non-mesh symmetry MOM capacitors

Non-shielding layer

```
x1 port1 port2 well momcaps_sy_mml110ae c=100f nf=20 nm=5 m=4 bm=1  
shielding_flag=0 presim_flag=1 metal_ring_flag=0 or 1  
or  
x1 port1 port2 well momcaps_sy_mml110ae l=3u nf=20 nm=5 m=4 bm=1 shielding_flag=0  
presim_flag=1 metal_ring_flag=0 or 1
```

With shielding layer

```
x1 port1 port2 well momcaps_sy_mml110ae c=100f nf=20 nm=5 m=4 bm=2  
shielding_flag=1 presim_flag=1 metal_ring_flag=0 or 1  
or  
x1 port1 port2 well momcaps_sy_mml110ae l=3u nf=20 nm=5 m=4 bm=2 shielding_flag=1  
presim_flag=1 metal_ring_flag=0 or 1
```

For mesh symmetry MOM capacitors

Non-shielding layer

```
x1 port1 port2 port3 port4 well momcaps_symesh_mml110ae c=100f mh=1 nf=20 nm=5  
m=4 bm=1 shielding_flag=0 presim_flag=1  
or  
x1 port1 port2 port3 port4 well momcaps_symesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4  
bm=1 shielding_flag=0 presim_flag=1
```

With shielding layer

```
x1 port1 port2 port3 port4 well momcaps_symesh_mml110ae c=100f mh=1 nf=20 nm=5  
m=4 bm=2 shielding_flag=1 presim_flag=1  
or  
x1 port1 port2 port3 port4 well momcaps_symesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4  
bm=2 shielding_flag=1 presim_flag=1
```

For non-mesh asymmetry MOM capacitors

```
x1 port1 port2 well momcaps_as_mml110ae c=100f nf=20 nm=5 m=4 bm=1  
metal_ring_flag=0 or 1 presim_flag=1  
or  
x1 port1 port2 well momcaps_as_mml110ae l=3u nf=20 nm=5 m=4 bm=1 metal_ring_flag=0  
or 1 presim_flag=1
```

For mesh asymmetry MOM capacitors

x1 port1 port2 port3 port4 well momcaps_asmesh_mml110ae c=100f mh=1 nf=20 nm=5
m=4 bm=1 presim_flag=1

or

x1 port1 port2 port3 port4 well momcaps_asmesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4
bm=1 presim_flag=1

For non-mesh stripe MOM capacitors

Non-shielding layer

x1 port1 port2 well momcaps_stripe_mml110ae c=100f nf=20 nm=5 m=4 bm=1
shielding_flag=0 presim_flag=1 metal_ring_flag=0 or 1

or

x1 port1 port2 well momcaps_stripe_mml110ae l=3u nf=20 nm=5 m=4 bm=1
shielding_flag=0 presim_flag=1 metal_ring_flag=0 or 1

With shielding layer

x1 port1 port2 well momcaps_stripe_mml110ae c=100f nf=20 nm=5 m=4 bm=2
shielding_flag=1 presim_flag=1 metal_ring_flag=0 or 1

or

x1 port1 port2 well momcaps_stripe_mml110ae l=3u nf=20 nm=5 m=4 bm=2
shielding_flag=1 presim_flag=1 metal_ring_flag=0 or 1

For mesh stripe MOM capacitors

Non-shielding layer

x1 port1 port2 port3 port4 well momcaps_stripemesh_mml110ae c=100f mh=1 nf=20 nm=5
m=4 bm=1 shielding_flag=0 presim_flag=1

or

x1 port1 port2 port3 port4 well momcaps_stripemesh_mml110ae l=3u mh=1 nf=20 nm=5
m=4 bm=1 shielding_flag=0 presim_flag=1

With shielding layer

x1 port1 port2 port3 port4 well momcaps_stripemesh_mml110ae c=100f mh=1 nf=20 nm=5
m=4 bm=2 shielding_flag=1 presim_flag=1

or

x1 port1 port2 port3 port4 well momcaps_stripemesh_mml110ae l=3u mh=1 nf=20 nm=5
m=4 bm=2 shielding_flag=1 presim_flag=1

3. For Mentor Graphics Eldo

-Copy L110AE_MOM_V021.lib.eldo & L110AE_MOM_V021.mdl.eldo into your simulation directory.

-Add the following statement in the input file to the simulator for 90% dimension shrinkage.
.Option scale=0.9

.lib "/L110AE_MOM_V021.lib.eldo" CornerSection

, where CornerSection could be tt, ss, ff.

-Follow the example below to call the model

For non-mesh symmetry MOM capacitors

Non-shielding layer

x1 port1 port2 well momcaps_sy_mml110ae c=100f nf=20 nm=5 m=4 bm=1

shielding_flag=0 presim_flag=1 metal_ring_flag=0 or 1

or

x1 port1 port2 well momcaps_sy_mml110ae l=3u nf=20 nm=5 m=4 bm=1 shielding_flag=0

presim_flag=1 metal_ring_flag=0 or 1

With shielding layer

x1 port1 port2 well momcaps_sy_mml110ae c=100f nf=20 nm=5 m=4 bm=2

shielding_flag=1 presim_flag=1 metal_ring_flag=0 or 1

or

x1 port1 port2 well momcaps_sy_mml110ae l=3u nf=20 nm=5 m=4 bm=2 shielding_flag=1

presim_flag=1 metal_ring_flag=0 or 1

For mesh symmetry MOM capacitors

Non-shielding layer

x1 port1 port2 port3 port4 well momcaps_symesh_mml110ae c=100f mh=1 nf=20 nm=5

m=4 bm=1 shielding_flag=0 presim_flag=1

or

x1 port1 port2 port3 port4 well momcaps_symesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4

bm=1 shielding_flag=0 presim_flag=1

With shielding layer

x1 port1 port2 port3 port4 well momcaps_symesh_mml110ae c=100f mh=1 nf=20 nm=5

m=4 bm=2 shielding_flag=1 presim_flag=1

or

x1 port1 port2 port3 port4 well momcaps_symesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4

bm=2 shielding_flag=1 presim_flag=1

For non-mesh asymmetry MOM capacitors

x1 port1 port2 well momcaps_as_mml110ae c=100f nf=20 nm=5 m=4 bm=1

metal_ring_flag=0 or 1 presim_flag=1

or

x1 port1 port2 well momcaps_as_mml110ae l=3u nf=20 nm=5 m=4 bm=1 metal_ring_flag=0

or 1 presim_flag=1

For mesh asymmetry MOM capacitors

x1 port1 port2 port3 port4 well momcaps_asmesh_mml110ae c=100f mh=1 nf=20 nm=5

m=4 bm=1 presim_flag=1

or

x1 port1 port2 port3 port4 well momcaps_asmesh_mml110ae l=3u mh=1 nf=20 nm=5 m=4

bm=1 presim_flag=1

For non-mesh stripe MOM capacitors

Non-shielding layer

x1 port1 port2 well momcaps_stripe_mml110ae c=100f nf=20 nm=5 m=4 bm=1
shielding_flag=0 presim_flag=1 metal_ring_flag=0 or 1
or
x1 port1 port2 well momcaps_stripe_mml110ae l=3u nf=20 nm=5 m=4 bm=1
shielding_flag=0 presim_flag=1 metal_ring_flag=0 or 1

With shielding layer

x1 port1 port2 well momcaps_stripe_mml110ae c=100f nf=20 nm=5 m=4 bm=2
shielding_flag=1 presim_flag=1 metal_ring_flag=0 or 1
or
x1 port1 port2 well momcaps_stripe_mml110ae l=3u nf=20 nm=5 m=4 bm=2
shielding_flag=1 presim_flag=1 metal_ring_flag=0 or 1

For mesh symmetry MOM capacitors

Non-shielding layer

x1 port1 port2 port3 port4 well momcaps_stripemesh_mml110ae c=100f mh=1 nf=20 nm=5
m=4 bm=1 shielding_flag=0 presim_flag=1
or
x1 port1 port2 port3 port4 well momcaps_stripemesh_mml110ae l=3u mh=1 nf=20 nm=5
m=4 bm=1 shielding_flag=0 presim_flag=1

*Note: The "mh" is number of unit cell and the "nm" is number of metal layers.
The "m" is number of device and the default value is "1".

ModelFileName: L110AE_MOM_V022

DeviceName for Symmetric type MOM Capacitor: momcaps_sy_mml110ae

DeviceName for Asymmetric type MOM Capacitor: momcaps_asy_mml110ae

DeviceName for Symmetric type Mesh MOM Capacitor: momcaps_symesh_mml110ae

DeviceName for Asymmetric type Mesh MOM Capacitor: momcaps_ashesh_mml110ae

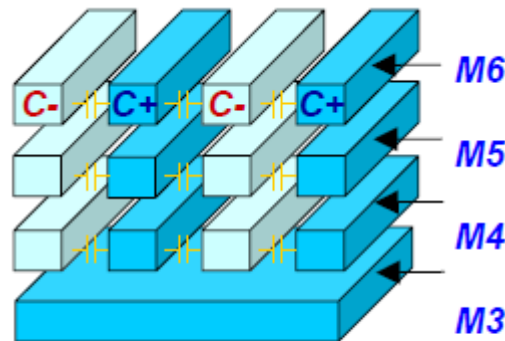
DeviceName for Stripe type Mesh MOM Capacitor: momcaps_stripemesh_mml110ae

Temperature range: -55 °C ~ +125 °C

1.4 Device Architecture and Layout

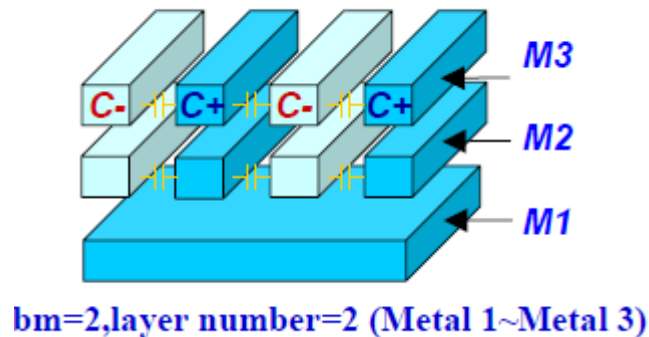
1. MOM capacitors rely on coupling capacitance between metal fingers running in parallel. Metal layers are stacked from Metal 1 to upper metal layer to increase the capacitance density.
2. The lowest metal layer is limited to Metal1. Metal finger width and spacing are fixed at the minimum dimensions, i.e. 0.2mm, to achieve the maximum capacitance density.
3. For symmetric type MOM structures, the architecture is basically symmetric in terms of the 2 ports. Number of fingers per layer is limited to even number to keep the symmetry. The minimum number of metal layers is 3 and maximum number of metal layer is 6.
4. For asymmetric type MOM capacitor, one port (bottom port) is connected to metal1 plate that shields the other port (top port) from the substrate, similarly to traditional planar MIM capacitor. Number of fingers per layer is limited to odd number. The minimum number of metal layers is 4 and maximum number of metal layer is 6.
5. Design parameters for MOM capacitors are finger length (l), number of fingers per layer (nf), number of metal layers (nm) and number of mesh (mh). The design was suggested to use square case to ensure the model accuracy.
6. For stripe type MOM without shielding layer structures, the architecture is basically symmetric in terms of the 2 ports. Number of fingers per layer is limited to even number to keep the symmetry. The minimum number of metal layers is 2 and maximum number of metal layer is 6. The bottom layer which can be selected is from 1 to 5.
7. For stripe type MOM with shielding layer structures, the architecture is basically symmetric in terms of the 2 ports. Number of fingers per layer is limited to even number to keep the symmetry. If $bm > 1$, the layer $bm-1$ will become shielding layer. The minimum number of metal layers is 2 and the maximum number of metal layer is 5. The bottom layer which can be selected is from 2 to 5.

Example: Sym MOMCAP, $bm=4$, layer number=3. Metal 3 is shielding layer



**$bm=4$, Layer number=3
(Metal3~Metal6)**

Example: Stripe MOMCAP, $bm=2$, layer number=2 Metal 1 is shielding layer



MOM Square case (Sym and Asym):

Pitch = metal width + spacing between metal to metal=0.2um

L is roughly equal to "NF * Pitch".

8. The guard ring (Ground ring) is an optional selection function for non-mesh MOMcap and is must for mesh MOMcap; the purpose is to reduce the parasitic coupling from outside environment.
9. We would suggest the spacing is larger than 0.8um to minimize mutual coupling between MOM body and guard ring if guard ring was selected. However, the spacing must fix with 0.8um while mesh MOMcap application for model accuracy concern.
10. The pickup ports (C+ and C-) for metal connection is suggested to use metal layer 2 or metal layer 3.
11. The metal routing above (top metal +1) layer of MOMcap is not allowed due to model accuracy concern from routing coupling effect.

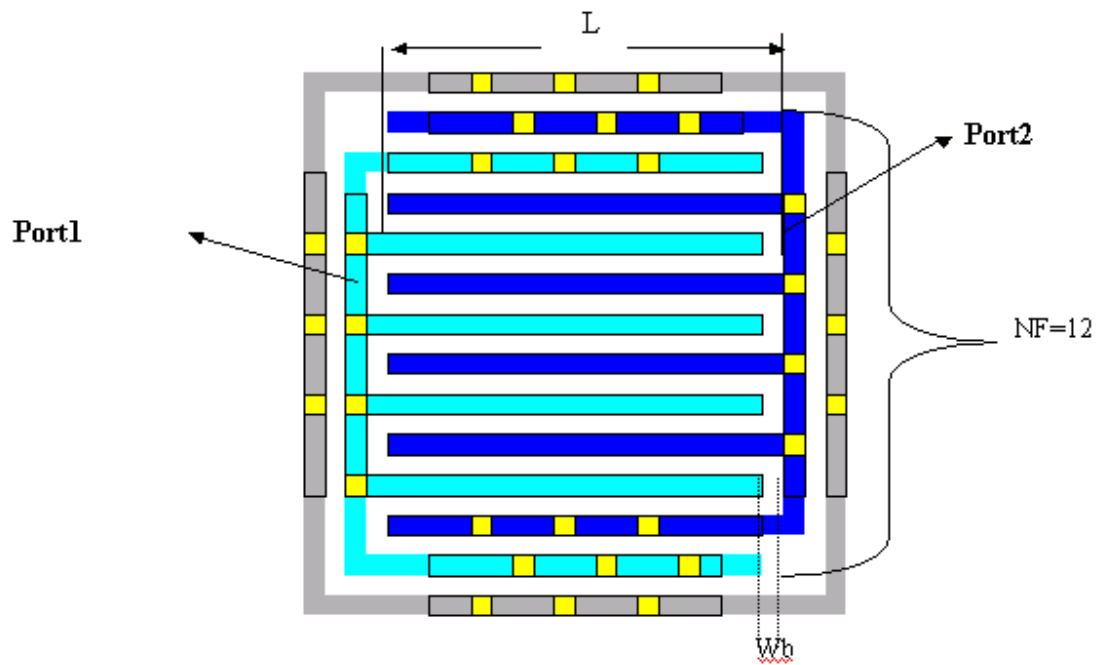


Figure 1-1 Example layout of Symmetric type MOM capacitor

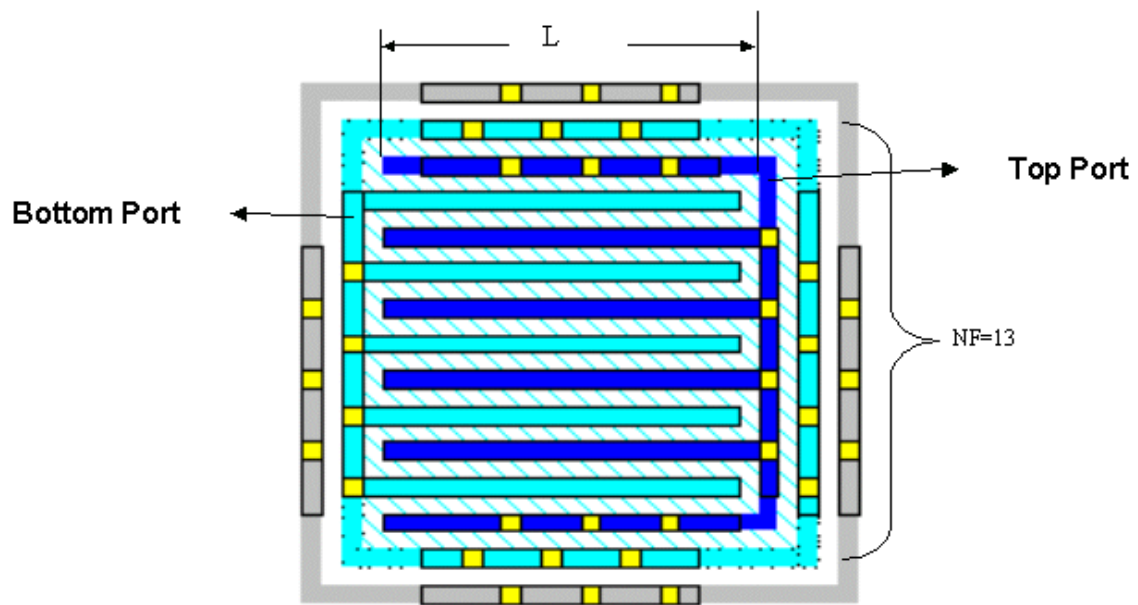


Figure 1-2 Example layout of Asymmetric type MOM capacitor

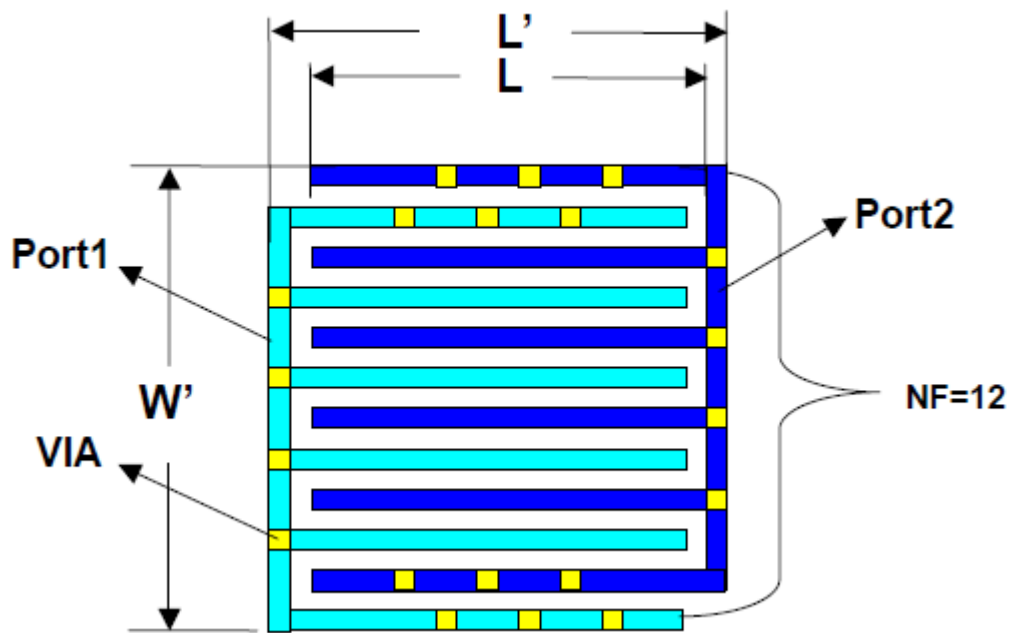


Figure 1-3 Example layout of Stripe type MOM capacitor

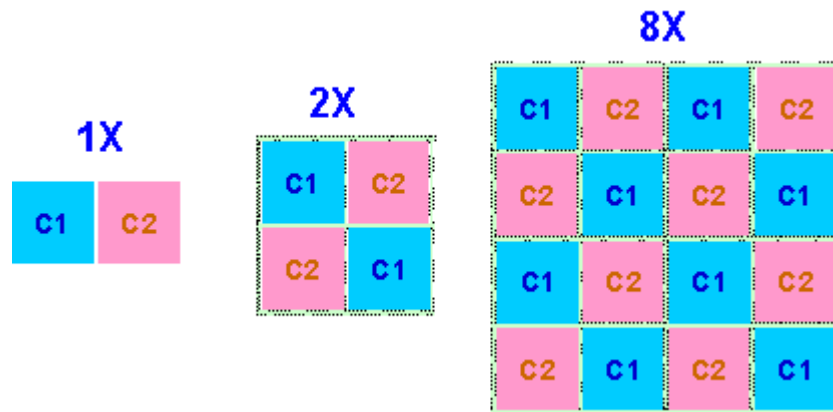


Figure 1-4 Top view of Mesh capacitor

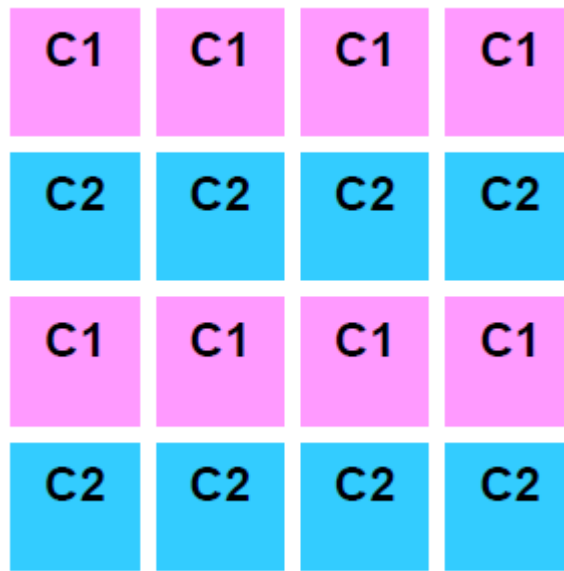


Figure 1-5 Top view of Stripe Mesh MOM capacitor

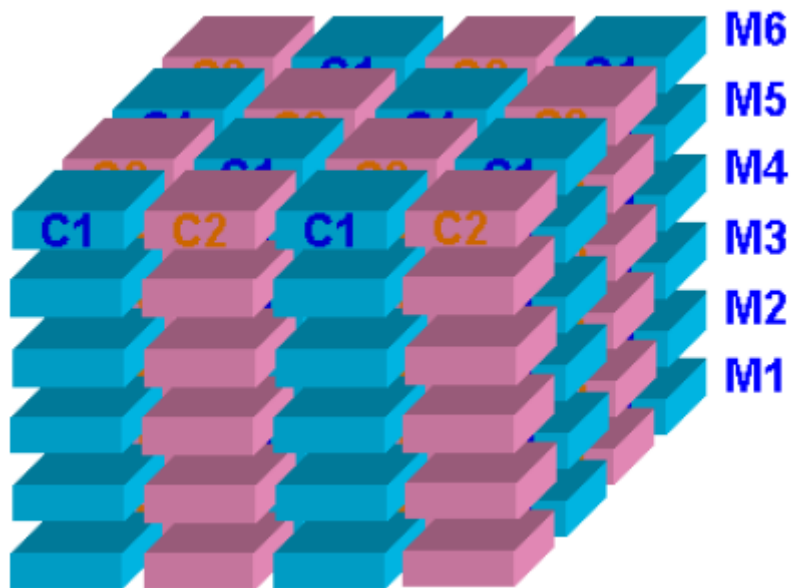


Figure 1-6 Top view of Stripe Mesh MOM capacitor

2 Sub-Circuit Model Topology

2.1 Equivalent Sub-Circuit of MOM Capacitor Model

a. Symmetric Type MOM Mixed-Mode Model

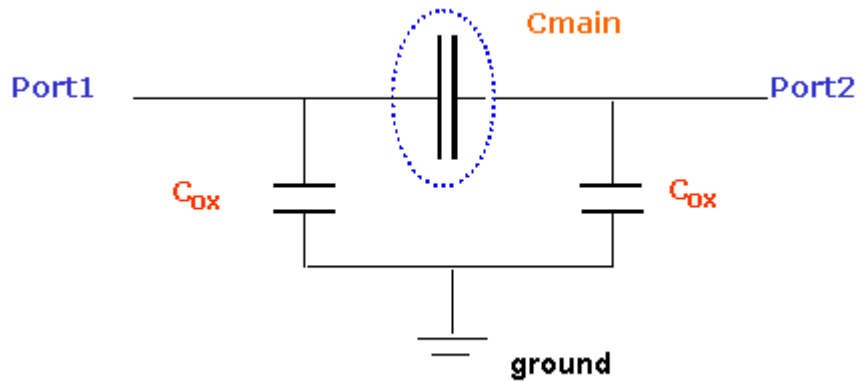


Figure 2-1 Sub-circuit model of symmetric type MOM capacitor

b. Asymmetric Type MOM Mixed-Mode Model

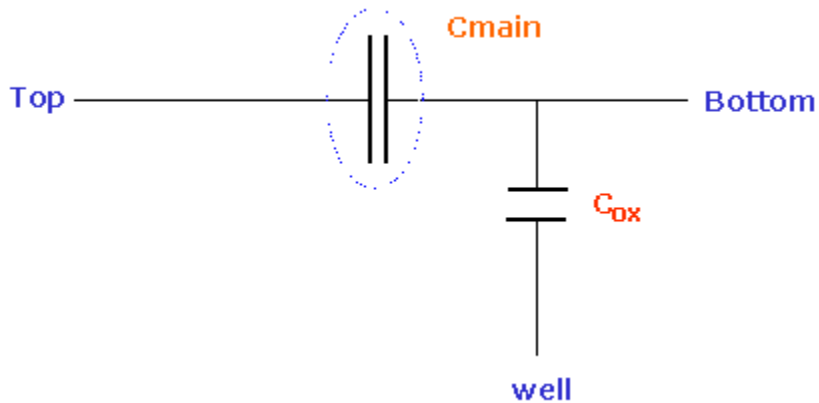


Figure 2-2 Sub-circuit model of asymmetric type MOM capacitor

Note: - The current version models cover only capacitance (main capacitance and substrate coupling capacitance), no other parasitic such as inductance and resistance are included in this model, therefore it is recommended to be used only for low frequency application.

2.2 Model equations and parameters

For Non-Mesh Symmetry MOM Capacitor

-To get capacitance value for given l (before shrinkage), nf, nm and bm

$$c = bm1 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb_bm1 + 2 * ratio_bm1 * cho_tb_bm1) + (nf - 2) * cv_tb_bm1) * factor) - 0.45f \\ + bm2 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor) - 0.45f \\ + bm3 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor) - 0.45f \\ + bm4 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor) - 0.45f$$

To get dimension (l) (before shrinkage) for given capacitance c, nf, nm and bm

$$l = bm1 * (((c + 0.45f) / factor - (nf - 2) * cv_tb_bm1) / ((nf - 3) * cho_tb_bm1 + 2 * ratio_bm1 * cho_tb_bm1)) * 1u / 0.9 \\ + bm2 * (((c + 0.45f) / factor - (nf - 2) * cv_tb_bm2) / ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2)) * 1u / 0.9 \\ + bm3 * (((c + 0.45f) / factor - (nf - 2) * cv_tb_bm2) / ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2)) * 1u / 0.9 \\ + bm4 * (((c + 0.45f) / factor - (nf - 2) * cv_tb_bm2) / ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2)) * 1u / 0.9$$

For Non-Mesh Asymmetry MOM Capacitor

-To get capacitance value for given l (before shrinkage), nf, nm and bm

$$c = bm1 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb + 2 * ratio * cho_tb) + (nf - 2) * cv_tb) * factor) - 0.45f \\ + bm2 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb + 2 * ratio * cho_tb) + (nf - 2) * cv_tb) * factor) - 0.45f \\ + bm3 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb + 2 * ratio * cho_tb) + (nf - 2) * cv_tb) * factor) - 0.45f$$

To get dimension (l) (before shrinkage) for given capacitance c, nf, nm and bm

$$l = bm1 * (((c + 0.45f) / factor - (nf - 2) * cv_tb) / ((nf - 3) * cho_tb + 2 * ratio * cho_tb)) * 1u / 0.9 \\ + bm2 * (((c + 0.45f) / factor - (nf - 2) * cv_tb) / ((nf - 3) * cho_tb + 2 * ratio * cho_tb)) * 1u / 0.9 \\ + bm3 * (((c + 0.45f) / factor - (nf - 2) * cv_tb) / ((nf - 3) * cho_tb + 2 * ratio * cho_tb)) * 1u / 0.9$$

For Non-Mesh Stripe MOM Capacitor

-To get capacitance value for given l (before shrinkage), nf, nm and bm

$$c = bm1 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb_bm1 + 2 * ratio_bm1 * cho_tb_bm1) + (nf - 2) * cv_tb_bm1) * factor) - 0.00184f * l * 0.9 - 0.03164f \\ + bm2 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor) - 0.00184f * l * 0.9 - 0.03164f \\ + bm3 * (((l * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor) - 0.00184f * l * 0.9 - 0.03164f$$

$$+bm4*(((l*0.9)/1u*((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2) + (nf-2)*cv_tb_bm2) * factor) -0.00184f*l*0.9-0.03164f)$$

$$+bm5*(((l*0.9)/1u*((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2) + (nf-2)*cv_tb_bm2) * factor) -0.00184f*l*0.9-0.03164f)$$

-To get dimension (l) (before shrinkage) for given capacitance c, nf, nm and bm

$$l= bm1*(((c+0.03164f)/factor-(nf-2)*cv_tb_bm1) / ((nf-3)*cho_tb_bm1 + 2*ratio_bm1*cho_tb_bm1-0.00184f/factor))*1u/0.9)$$

$$+bm2*(((c+0.03164f)/factor-(nf-2)*cv_tb_bm2) / ((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2-0.00184f/factor))*1u/0.9)$$

$$+bm3*(((c+0.03164f)/factor-(nf-2)*cv_tb_bm2) / ((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2-0.00184f/factor))*1u/0.9)$$

$$+bm4*(((c+0.03164f)/factor-(nf-2)*cv_tb_bm2) / ((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2-0.00184f/factor))*1u/0.9)$$

$$+bm5*(((c+0.03164f)/factor-(nf-2)*cv_tb_bm2) / ((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2-0.00184f/factor))*1u/0.9)$$

For Symmetry Mesh MOM Capacitor

-To get capacitance value for given l2 (before shrinkage) (the finger length of unit cell capacitor), nf, nm and bm

$$c2= bm1*(((l2*0.9)/1u*((nf-3)*cho_tb_bm1 + 2*ratio_bm1*cho_tb_bm1) + (nf-2)*cv_tb_bm1) * factor))$$

$$+bm2*(((l2*0.9)/1u*((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2) + (nf-2)*cv_tb_bm2) * factor))$$

$$+bm3*(((l2*0.9)/1u*((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2) + (nf-2)*cv_tb_bm2) * factor))$$

$$+bm4*(((l2*0.9)/1u*((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2) + (nf-2)*cv_tb_bm2) * factor))$$

, where c2 is the capacitance of unit cell capacitor.

$c=c2*mh$

To get dimension (l2 (before shrinkage), l2 is the finger length of unit cell capacitor) for given capacitance c, nf, nm and bm

$$l2= bm1*(((c2/factor-(nf-2)*cv_tb_bm1) / ((nf-3)*cho_tb_bm1 + 2*ratio_bm1*cho_tb_bm1))*1u/0.9)$$

$$+bm2*(((c2/factor-(nf-2)*cv_tb_bm2) / ((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2))*1u/0.9)$$

$$+bm3*(((c2/factor-(nf-2)*cv_tb_bm2) / ((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2))*1u/0.9)$$

$$+bm4*(((c2/factor-(nf-2)*cv_tb_bm2) / ((nf-3)*cho_tb_bm2 + 2*ratio_bm2*cho_tb_bm2))*1u/0.9)$$

, where $c2=c/mh$

For Asymmetry Mesh MOM Capacitor

-To get capacitance value for given l2 (before shrinkage) (the finger length of unit cell capacitor), nf, nm and bm

$$c2 = bm1 * (((l2 * 0.9) / 1u * ((nf - 3) * cho_tb + 2 * ratio * cho_tb) + (nf - 2) * cv_tb) * factor)) \\ + bm2 * (((l2 * 0.9) / 1u * ((nf - 3) * cho_tb + 2 * ratio * cho_tb) + (nf - 2) * cv_tb) * factor)) \\ + bm3 * (((l2 * 0.9) / 1u * ((nf - 3) * cho_tb + 2 * ratio * cho_tb) + (nf - 2) * cv_tb) * factor))$$

, where c2 is the capacitance of unit cell capacitor.

$$c = c2 * mh$$

To get dimension (l2 (before shrinkage), l2 is the finger length of unit cell capacitor) for given capacitance c, nf, nm and bm

$$l2 = bm1 * (((c2 / factor - (nf - 2) * cv_tb) / ((nf - 3) * cho_tb + 2 * ratio * cho_tb)) * 1u / 0.9) \\ + bm2 * (((c2 / factor - (nf - 2) * cv_tb) / ((nf - 3) * cho_tb + 2 * ratio * cho_tb)) * 1u / 0.9) \\ + bm3 * (((c2 / factor - (nf - 2) * cv_tb) / ((nf - 3) * cho_tb + 2 * ratio * cho_tb)) * 1u / 0.9)$$

, where c2=c/mh

For Stripe Mesh MOM Capacitor

-To get capacitance value for given l2 (before shrinkage) (the finger length of unit cell capacitor), nf, nm and bm

$$c2 = bm1 * (((l2 * 0.9) / 1u * ((nf - 3) * cho_tb_bm1 + 2 * ratio_bm1 * cho_tb_bm1) + (nf - 2) * cv_tb_bm1) * factor)) \\ + bm2 * (((l2 * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor)) \\ + bm3 * (((l2 * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor)) \\ + bm4 * (((l2 * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor)) \\ + bm5 * (((l2 * 0.9) / 1u * ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2) + (nf - 2) * cv_tb_bm2) * factor))$$

, where c2 is the capacitance of unit cell capacitor.

$$c = c2 * mh$$

To get dimension (l2 (before shrinkage), l2 is the finger length of unit cell capacitor) for given capacitance c, nf, nm and bm

$$l2 = bm1 * (((c2) / factor - (nf - 2) * cv_tb_bm1) / ((nf - 3) * cho_tb_bm1 + 2 * ratio_bm1 * cho_tb_bm1)) * 1u / 0.9) \\ + bm2 * (((c2) / factor - (nf - 2) * cv_tb_bm2) / ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2)) * 1u / 0.9) \\ + bm3 * (((c2) / factor - (nf - 2) * cv_tb_bm2) / ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2)) * 1u / 0.9) \\ + bm4 * (((c2) / factor - (nf - 2) * cv_tb_bm2) / ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2)) * 1u / 0.9) \\ + bm5 * (((c2) / factor - (nf - 2) * cv_tb_bm2) / ((nf - 3) * cho_tb_bm2 + 2 * ratio_bm2 * cho_tb_bm2)) * 1u / 0.9)$$

, where c2=c/mh

- The unit for l is meter, for c is Farad
- Factor is a scaling parameter related to number of metal layer (nm)
- Model parameters are extracted based on silicon fabricated by UMC's Fab.12A.
- The temperatures coefficient is extracted by 55nm's silicon.
- The process corners (min, typ, max) with +/- 20% variation on main capacitance are implemented in the spice model

Table 2-1 Model parameters

Parameters			Symmetric type
cho_tb_bm1			3.24e-16
cv_tb_bm1			4.1e-17
ratio_bm1			1.2
cho_tb_bm2			3.45e-16
cv_tb_bm2			1e-17
ratio_bm2			1.27
factor	bm=1	nm=6	2.058
		nm=5	1.69
		nm=4	1.35
		nm=3	1
	bm=2	nm=5	1.67
		nm=4	1.32
		nm=3	1
	bm=3	nm=4	1.32
		nm=3	1
	bm=4	nm=3	1
Temperature coefficient		tc1	-3.098e-5
		tc2	-1.122e-9
Voltage coefficient		vc1	2.874e-7
		vc2	-1.089e-6

Parameters			Symmetric type
cho_tb			3.24e-16
cv_tb			4.1e-17
ratio			1.2
factor	bm=1	nm=6	1
		nm=5	0.8
		nm=4	0.598
	bm=2	nm=5	0.8
		nm=4	0.598
	bm=3	nm=4	0.598
Temperature coefficient		tc1	-3.098e-5
		tc2	-1.122e-9
Voltage coefficient		vc1	2.874e-7
		vc2	-1.089e-6

Parameters			Symmetric type
cho_tb_bm1			6.63e-16
cv_tb_bm1			2e-17
ratio_bm1			1.44
cho_tb_bm2			2.24e-16
cv_tb_bm2			2e-17
ratio_bm2			1.3
factor	bm=1	nm=6	1
		nm=5	0.8262
		nm=4	0.651
		nm=3	0.4812
		nm=2	0.310
	bm=2	nm=5	2.5
		nm=4	2.01
		nm=3	1.5028
		nm=2	1
	bm=3	nm=4	2.01
		nm=3	1.5028
		nm=2	1
	bm=4	nm=3	1.5028
		nm=2	1
	bm=5	nm=2	1
Temperature coefficient		tc1	-3.098e-5
		tc2	-1.122e-9
Voltage coefficient		vc1	2.874e-7
		vc2	-1.089e-6

2.3 Model Scaling Range

Table 2-2 Model scaling range for Symmetric MOM capacitors

Parameter		Unit	Min	Max	Limitation
l		um	2.4	78.4	
nf		integer	8	198	Even number
nm		integer	3	6	
w		um	Fixed at 0.2		
s		um	Fixed at 0.2		
c	nm=6	F	10.71f	9.3p	Bm=1
	nm=5	F	8.72f	7.64p	
	nm=4	F	6.87f	6.1p	
	nm=3	F	4.97f	4.52p	

Table 2-3 Model scaling range for Asymmetric MOM capacitors

Parameter		Unit	Min	Max	Limitation
l		um	2.4	78.4	
nf		integer	9	199	odd number
nm		integer	4	6	
w		um	Fixed at 0.2		
s		um	Fixed at 0.2		
c	nm=6	F	12.18f	8.21p	Bm=1
	nm=5	F	9.66f	6.56p	
	nm=4	F	7.11f	4.91p	

Table 2-4 Model scaling range for stripe MOM capacitors

Parameter		Unit	Min	Max	Limitation
l		um	2.52	78.4	
nf		integer	8	198	odd number
nm		integer	2	6	
w		um	Fixed at 0.2		
s		um	Fixed at 0.2		
c	nm=6	F	11.94f	9.26p	bm=1
	nm=5	F	9.86f	7.65p	
	nm=4	F	7.76f	6.03p	
	nm=3	F	5.73f	4.46p	
	nm=2	F	3.68f	2.87p	

2.4 Model Accuracy Verification

Table 2-5 Simulation results of Symmetric type without shielding layer MOM capacitor Model

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for non-shielding and metal_ring_flag=1		
					Cmain	As-Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	3	2.52	8	1~3	5.733891367	5.24	5.23	0.74049
256	3	2.52	8	1~3	1467.87619	1456.60	1455.00	189.57
1	3	4.4	14	1~3	17.75179145	17.25	17.24	1.6719
4	3	4.4	14	1~3	71.00716578	70.79	70.74	6.6874
64	3	4.4	14	1~3	1136.114653	1132.60	1131.80	107
256	3	4.4	14	1~3	4544.45861	4530.40	4527.30	427.99
1	3	9.5	28	1~3	77.27464328	76.55	76.52	4.6018
4	3	9.5	28	1~3	309.0985731	307.98	307.88	18.407
64	3	9.5	28	1~3	4945.57717	4927.70	4926.00	294.52
1	3	28.5	72	1~3	603.4171638	595.87	595.80	21.209
4	3	28.5	72	1~3	2413.668655	2385.30	2385.00	84.836
16	3	28.5	72	1~3	9654.67462	9541.10	9539.90	339.34
1	3	78.4	198	1~3	4573.25006	4520.60	4520.40	119.2
1	4	9.5	28	1~4	102.579845	103.49	103.46	5.3514
1	4	28.5	72	1~4	804.16177	804.58	804.48	23.377
1	4	78.4	198	1~4	6190.36454	6103.00	6102.70	125.26
1	5	9.5	28	1~5	133.214258	130.60	129.63	6.0676
4	5	9.5	28	1~5	532.8570322	524.19	520.31	24.27
64	5	9.5	28	1~5	8525.712515	8387.00	8325.00	388.32
1	5	28.5	72	1~5	1023.411865	1014.50	1007.20	25.45
1	5	78.4	198	1~5	7843.49509	7694.40	7639.80	131.06
1	6	9.5	28	1~6	159.366325	158.47	157.95	6.80
1	6	28.5	72	1~6	1247.378325	1230.30	1226.60	27.57
1	6	78.4	198	1~6	9581.7954	9331.10	9303.50	136.99
1	6	2.4	14	1~6	20.44948317	19.95	19.86	1.97
1	6	2.4	102	1~6	152.3450404	154.69	154.03	13.40
1	6	2.4	198	1~6	296.328925	301.68	300.40	25.87
1	6	4.8	102	1~6	297.7836646	301.16	300.07	15.50
1	6	20	8	1~6	90.02751855	89.15	88.87	6.49
1	6	20	198	1~6	2403.451639	2392.70	2385.30	51.61
1	6	40	14	1~6	323.0658256	323.19	322.22	14.67

1	6	78.4	8	1~6	351.6226224	349.25	348.22	24.08
1	6	78.4	14	1~6	632.5224879	632.88	631.02	27.64
1	6	78.4	102	1~6	4837.456906	4792.90	4778.70	79.94
1	5	2.4	14	1~5	16.88992327	16.37	16.23	1.73
1	5	2.4	102	1~5	125.5663597	127.48	126.41	11.82
1	5	2.4	198	1~5	244.3774574	248.69	246.60	22.83
1	5	4.8	102	1~5	245.7761729	248.26	246.34	13.83
1	5	20	8	1~5	74.29185052	73.43	72.90	5.67
1	5	20	198	1~5	1976.938664	1973.00	1958.70	47.90
1	5	40	14	1~5	266.2124614	266.43	264.52	13.00
1	5	78.4	8	1~5	289.6146424	287.91	285.87	21.04
1	5	78.4	14	1~5	521.1056102	521.80	518.10	24.51
1	5	78.4	102	1~5	3981.649851	3952.20	3924.20	75.47
1	3	2.4	14	1~3	9.95461821	9.43	9.42	1.22
1	3	2.4	102	1~3	73.78682949	74.71	74.61	8.65
1	3	2.4	198	1~3	143.608313	145.93	145.73	16.74
1	3	4.8	102	1~3	144.3764683	145.68	145.58	10.47
1	3	20	8	1~3	43.05293722	42.96	42.95	4.01
1	3	20	198	1~3	1157.541282	1159.00	1158.80	40.47
1	3	40	14	1~3	154.6196712	156.35	156.34	9.64
1	3	78.4	8	1~3	166.9208034	168.98	168.97	14.95
1	3	78.4	14	1~3	302.674774	306.40	306.39	18.24
1	3	78.4	102	1~3	2325.415386	2321.90	2321.80	66.53

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for non-shielding and metal_ring_flag=0		
					Cmain	As-Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	3	2.52	8	1~3	5.733891367	5.2398	5.2338	0.55992
1	3	4.4	14	1~3	17.75179145	17.247	17.235	1.2653
1	3	9.5	28	1~3	77.27464328	76.545	76.519	3.6303
1	3	28.5	72	1~3	603.4171638	595.87	595.8	18.305
1	3	78.4	198	1~3	4573.25006	4520.6	4520.4	111
1	4	9.5	28	1~4	102.579845	103.49	103.46	4.0614
1	4	28.5	72	1~4	804.16177	804.58	804.48	19.521
1	4	78.4	198	1~4	6190.36454	6103	6102.7	114.37
1	5	9.5	28	1~5	133.214258	130.6	129.63	4.446
1	5	28.5	72	1~5	1023.411865	1014.5	1007.2	20.606
1	5	78.4	198	1~5	7843.49509	7694.4	7639.8	117.37

1	6	9.5	28	1~6	159.366325	158.47	157.95	4.796
1	6	28.5	72	1~6	1247.378325	1230.3	1226.6	21.593
1	6	78.4	198	1~6	9581.7954	9331.1	9303.5	120.11
1	6	2.4	14	1~6	20.44948317	19.946	19.862	1.3523
1	6	2.4	102	1~6	152.3450404	154.69	154.03	8.9573
1	6	2.4	198	1~6	296.328925	301.68	300.4	17.254
1	6	4.8	102	1~6	297.7836646	301.16	300.07	10.798
1	6	20	8	1~6	90.02751855	89.146	88.873	4.2203
1	6	20	198	1~6	2403.451639	2392.7	2385.3	41.073
1	6	40	14	1~6	323.0658256	323.19	322.22	9.9655
1	6	78.4	8	1~6	351.6226224	349.25	348.22	15.457
1	6	78.4	14	1~6	632.5224879	632.88	631.02	18.762
1	6	78.4	102	1~6	4837.456906	4792.9	4778.7	67.232
1	5	2.4	14	1~5	16.88992327	16.369	16.23	1.2234
1	5	2.4	102	1~5	125.5663597	127.48	126.41	8.2155
1	5	2.4	198	1~5	244.3774574	248.69	246.6	15.843
1	5	4.8	102	1~5	245.7761729	248.26	246.34	10.014
1	5	20	8	1~5	74.29185052	73.432	72.901	3.8267
1	5	20	198	1~5	1976.938664	1973	1958.7	39.356
1	5	40	14	1~5	266.2124614	266.43	264.52	9.1819
1	5	78.4	8	1~5	289.6146424	287.91	285.87	14.047
1	5	78.4	14	1~5	521.1056102	521.8	518.1	17.31
1	5	78.4	102	1~5	3981.649851	3952.2	3924.2	65.167
1	3	2.4	14	1~3	9.95461821	9.4319	9.4199	0.92313
1	3	2.4	102	1~3	73.78682949	74.714	74.614	6.4868
1	3	2.4	198	1~3	143.608313	145.93	145.73	12.556
1	3	4.8	102	1~3	144.3764683	145.68	145.58	8.1879
1	3	20	8	1~3	43.05293722	42.959	42.953	2.9096
1	3	20	198	1~3	1157.541282	1159	1158.8	35.355
1	3	40	14	1~3	154.6196712	156.35	156.34	7.3559
1	3	78.4	8	1~3	166.9208034	168.98	168.97	10.76
1	3	78.4	14	1~3	302.674774	306.4	306.39	13.925
1	3	78.4	102	1~3	2325.415386	2321.9	2321.8	60.354

Table 2-6 Simulation results of Symmetric type with shielding layer MOM capacitor Model

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for shielding and metal_ring_flag=1		
					Cmain	As- Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	3	2.52	8	4~6	6.103293965	5.5439	5.5097	1.0557
256	3	2.52	8	4~6	1562.443255	1534.5	1525.7	270.25
1	3	4.4	14	4~6	18.99611818	18.276	18.168	2.5019
4	3	4.4	14	4~6	75.98447273	74.902	74.473	10.01
64	3	4.4	14	4~6	1215.751564	1198.4	1191.6	160.12
256	3	4.4	14	4~6	4863.006255	4793.7	4766.3	640.50
1	3	9.5	28	4~6	83.05582578	81.517	81.046	7.6597
4	3	9.5	28	4~6	332.2233031	327.87	325.98	30.64
64	3	9.5	28	4~6	5315.57285	5245.9	5215.8	490.22
1	3	28.5	72	4~6	649.7334691	637	633.33	42.091
4	3	28.5	72	4~6	2598.933876	2549.8	2535.1	168.36
16	3	28.5	72	4~6	10395.73551	10199	10140	673.46
1	3	78.4	198	4~6	4923.770415	4838.1	4810.3	269.79
1	4	9.5	28	3~6	106.98067	107.75	107.12	8.4258
1	4	28.5	72	3~6	841.493245	840.98	836.13	44.298
1	4	78.4	198	3~6	6495.772885	6386.5	6349.7	275.91
1	5	9.5	28	2~6	137.321371	137.25	135.65	9.1231
4	5	9.5	28	2~6	549.2854841	550.82	544.39	36.49
64	5	9.5	28	2~6	8788.567745	8813.1	8710.3	583.88
1	5	28.5	72	2~6	1060.802885	1070.5	1058	46.309
1	5	78.4	198	2~6	8152.02785	8128.4	8033.4	281.51
1	3	4.4	14	2~4	18.34549541	18.276	18.168	2.4491
256	3	4.4	14	2~4	4696.446825	4793.7	4766.3	626.97
1	3	9.5	28	2~4	78.089675	81.517	81.046	7.5324
1	3	28.5	72	2~4	613.729825	637	633.33	41.703
1	3	78.4	198	2~4	4734.570915	4838.1	4810.3	268.64

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for shielding and metal_ring_flag=0		
					Cmain	As-Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	3	2.52	8	4~6	6.103294	5.5439	5.5097	0.86452
1	3	4.4	14	4~6	18.996118	18.276	18.168	2.0716
1	3	9.5	28	4~6	83.055826	81.517	81.046	6.6313
1	3	28.5	72	4~6	649.73347	637	633.33	39.017
1	3	78.4	198	4~6	4923.7704	4838.1	4810.3	261.11
1	4	9.5	28	3~6	106.98067	107.75	107.12	7.0946
1	4	28.5	72	3~6	841.49325	840.98	836.13	40.318
1	4	78.4	198	3~6	6495.7729	6386.5	6349.7	264.68
1	5	9.5	28	2~6	137.32137	137.25	135.65	7.507
1	5	28.5	72	2~6	1060.8029	1070.5	1058	41.478
1	5	78.4	198	2~6	8152.0279	8128.4	8033.4	267.87
1	3	4.4	14	2~4	18.345495	18.276	18.168	2.0699
1	3	9.5	28	2~4	78.089675	81.517	81.046	6.6263
1	3	28.5	72	2~4	613.72983	637	633.33	38.994
1	3	78.4	198	2~4	4734.5709	4838.1	4810.3	260.99

Table 2-7 Simulation results of Asymmetric type MOM capacitor Model

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for metal_ring_flag=0		
					Cmain	As-Extracted(fF)	Centering(fF)	Cox (per port)(fF)
256	4	2.52	9	1~4	1983.850115	2019.50	2014.90	440.83
1	4	4.4	15	1~4	22.37295219	21.78	21.70	2.22
4	4	4.4	15	1~4	89.49180875	88.90	88.60	13.28
64	4	4.4	15	1~4	1431.86894	1422.40	1417.70	212.41
256	4	4.4	15	1~4	5727.47576	5689.70	5670.70	849.65
1	4	9.5	29	1~4	91.10073391	89.22	88.84	5.13
4	4	9.5	29	1~4	364.4029356	358.68	357.18	30.27
64	4	9.5	29	1~4	5830.44697	5738.90	5714.80	484.35
1	4	20	53	1~4	338.982625	338.41	336.84	11.59
1	4	28.5	73	1~4	665.490865	661.58	658.43	17.86
1	4	40	103	1~4	1320.327085	1306.80	1300.40	28.30
1	4	56.8	145	1~4	2642.82468	2606.80	2594.00	46.31
1	4	78.4	199	1~4	5020.060275	4931.20	4906.60	74.78
4	5	4.4	15	1~5	120.0867719	118.93	118.53	16.02

64	5	4.4	15	1~5	1921.38835	1902.90	1896.50	256.35
256	5	4.4	15	1~5	7685.5534	7611.70	7586.20	1025.40
1	5	78.4	199	1~5	6701.1318	6597.10	6564.20	80.74
4	6	4.4	15	1~6	150.4311066	148.67	148.17	18.88
64	6	4.4	15	1~6	2406.897705	2378.60	2370.70	302.05
256	6	4.4	15	1~6	9627.59082	9514.60	9482.70	1208.20
1	6	78.4	199	1~6	8402.06374	8246.50	8205.3	86.17
1	6	2.4	15	1~6	21.89252598	21.60	21.5	2.17
1	6	2.4	103	1~6	158.0388776	159.19	158.6	13.21
1	6	2.4	199	1~6	306.9679716	309.30	308.0	25.26
1	6	4.8	103	1~6	286.7101978	288.53	287.3	14.55
1	6	20	9	1~6	90.09781181	89.68	89.4	6.94
1	6	20	199	1~6	2161.506299	2147.40	2136.9	39.37
1	6	40	15	1~6	305.3120045	305.87	304.7	14.55
1	6	78.4	9	1~6	347.7021804	346.73	345.8	25.26
1	6	78.4	15	1~6	595.2532468	596.20	593.98	27.19
1	6	78.4	103	1~6	4296.872726	4255.00	4234.20	55.40
1	5	2.4	15	1~5	17.60479079	17.19	17.13	1.95
1	5	2.4	103	1~5	126.5078531	127.26	126.75	11.77
1	5	2.4	199	1~5	245.3362435	247.35	246.34	22.48
1	5	4.8	103	1~5	229.0072863	230.74	229.72	13.02
1	5	20	9	1~5	72.0390216	71.66	71.46	6.19
1	5	20	199	1~5	1722.181064	1717.80	1709.40	35.97
1	5	40	15	1~5	243.5640218	244.61	243.70	13.02
1	5	78.4	9	1~5	277.4465745	277.29	276.54	22.48
1	5	78.4	15	1~5	474.465236	476.87	475.09	24.32
1	5	78.4	103	1~5	3421.598135	3403.90	3387.30	51.30

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for metal_ring_flag=1		
					Cmain	As- Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	4	4.4	15	1~4	22.37295219	21.78	21.70	3.32
1	4	9.5	29	1~4	91.10073391	89.22	88.84	7.57
1	4	20	53	1~4	338.982625	338.41	336.84	16.55
1	4	28.5	73	1~4	665.490865	661.58	658.43	24.88
1	4	40	103	1~4	1320.327085	1306.80	1300.40	38.27
1	4	56.8	145	1~4	2642.82468	2606.80	2594.00	60.48
1	4	78.4	199	1~4	5020.060275	4931.20	4906.60	94.36

1	5	78.4	199	1~5	6701.1318	6597.10	6564.20	107.26
1	6	78.4	199	1~6	8402.06374	8246.50	8205.30	120.67
1	6	2.4	15	1~6	21.89252598	21.60	21.53	3.62
1	6	2.4	103	1~6	158.0388776	159.19	158.55	22.44
1	6	2.4	199	1~6	306.9679716	309.30	308.04	42.97
1	6	4.8	103	1~6	286.7101978	288.53	287.26	24.30
1	6	20	9	1~6	90.09781181	89.68	89.44	11.75
1	6	20	199	1~6	2161.506299	2147.40	2136.90	60.97
1	6	40	15	1~6	305.3120045	305.87	304.74	24.30
1	6	78.4	9	1~6	347.7021804	346.73	345.78	42.97
1	6	78.4	15	1~6	595.2532468	596.20	593.98	45.43
1	6	78.4	103	1~6	4296.872726	4255.00	4234.20	81.41
1	5	2.4	15	1~5	17.60479079	17.19	17.13	3.08
1	5	2.4	103	1~5	126.5078531	127.26	126.75	18.87
1	5	2.4	199	1~5	245.3362435	247.35	246.34	36.10
1	5	4.8	103	1~5	229.0072863	230.74	229.72	20.53
1	5	20	9	1~5	72.0390216	71.66	71.46	9.90
1	5	20	199	1~5	1722.181064	1717.80	1709.40	52.58
1	5	40	15	1~5	243.5640218	244.61	243.70	20.53
1	5	78.4	9	1~5	277.4465745	277.29	276.54	36.10
1	5	78.4	15	1~5	474.465236	476.87	475.09	38.35
1	5	78.4	103	1~5	3421.598135	3403.90	3387.30	71.30

Table 2-8 Simulation results of Stripe type without shielding layer MOM capacitor Model

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for non-shielding and metal_ring_flag=1		
					Cmain	As- Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	5	2.52	8	2~6	10.02	10.07	9.92	0.94
256	5	2.52	8	2~6	2565.21	2586.00	2547.90	240.09
1	5	4.4	14	2~6	31.01	31.15	30.73	2.03
8	5	4.4	14	2~6	248.06	249.49	246.07	16.21
64	5	4.4	14	2~6	1984.50	1995.90	1968.60	129.72
256	5	4.4	14	2~6	7938.01	7983.60	7874.40	518.86
1	5	9.5	28	2~6	133.99	135.18	133.42	5.17
64	5	9.5	28	2~6	8575.61	8653.30	8540.70	330.99
1	5	28.5	72	2~6	1035.64	1045.10	1031.90	20.56

8	5	28.5	72	2~6	8285.11	8361.10	8255.70	164.49
1	5	78.4	198	2~6	7838.32	7916.10	7817.70	99.70
1	4	2.52	8	3~6	8.10	8.07	7.97	0.72
8	4	2.52	8	3~6	64.79	64.78	64.02	5.74
256	4	2.52	8	3~6	2073.41	2072.90	2048.50	183.55
1	4	4.4	14	3~6	24.99	24.97	24.70	1.57
2	4	4.4	14	3~6	49.99	50.00	49.46	3.13
8	4	4.4	14	3~6	199.95	199.99	197.84	12.53
64	4	4.4	14	3~6	1599.61	1599.90	1582.80	100.25
256	4	4.4	14	3~6	6398.44	6399.60	6331.00	401.01
1	4	9.5	28	3~6	108.09	108.35	107.26	4.02
8	4	9.5	28	3~6	864.69	867.05	858.34	32.14
64	4	9.5	28	3~6	6917.51	6936.40	6866.70	257.10
1	4	28.5	72	3~6	837.18	837.74	829.67	15.99
8	4	28.5	72	3~6	6697.41	6702.10	6637.60	127.93
1	4	78.4	198	3~6	6397.13	6345.50	6285.40	77.48
1	2	2.52	8	5~6	4.02	3.98	3.95	0.39
8	2	2.52	8	5~6	32.12	32.07	31.85	3.10
256	2	2.52	8	5~6	1027.98	1026.20	1019.10	99.10
1	2	4.4	14	5~6	12.42	12.34	12.27	0.85
2	2	4.4	14	5~6	24.85	24.75	24.61	1.70
8	2	4.4	14	5~6	99.38	99.00	98.43	6.79
64	2	4.4	14	5~6	795.05	792.03	787.44	54.32
256	2	4.4	14	5~6	3180.19	3168.10	3149.80	217.28
1	2	9.5	28	5~6	54.14	53.62	53.35	2.25
2	2	9.5	28	5~6	108.29	107.31	106.76	4.50
8	2	9.5	28	5~6	433.16	429.23	427.04	17.98
64	2	9.5	28	5~6	3465.28	3433.90	3416.30	143.87
1	2	28.5	72	5~6	421.08	414.71	412.75	9.73
8	2	28.5	72	5~6	3368.60	3317.90	3302.30	77.82
1	2	78.4	198	5~6	3219.16	3141.30	3127.00	51.73
256	4	4.4	14	2~5	6318.65	6399.60	6331.00	436.43
64	4	9.5	28	2~5	6820.54	6936.40	6866.70	284.38
8	4	28.5	72	2~5	6582.94	6702.10	6637.60	147.55
1	4	78.4	198	2~5	6230.20	6345.50	6285.40	93.78
256	2	4.4	14	2~3	3032.84	3168.10	3149.80	282.19
64	2	9.5	28	2~3	3294.05	3433.90	3416.30	193.83
8	2	28.5	72	2~3	3192.23	3317.90	3302.30	114.00

1	2	78.4	198	2~3	3029.95	3141.30	3127.00	81.97
1	6	2.52	8	1~6	11.95	11.99	11.94	1.23
256	6	2.52	8	1~6	3058.55	3076.40	3064.10	314.51
1	6	4.4	14	1~6	36.82	36.76	36.65	2.65
2	6	4.4	14	1~6	73.64	73.58	73.36	5.29
8	6	4.4	14	1~6	294.55	294.31	293.45	21.179
64	6	4.4	14	1~6	2356.37	2354.5	2347.6	169.43
256	6	4.4	14	1~6	9425.49	9418	9390.5	677.72
1	6	9.5	28	1~6	159.08	158.88	158.53	6.80
8	6	9.5	28	1~6	1272.67	1271.3	1268.5	54.367
64	6	9.5	28	1~6	10181.39	10170	10148	434.94
1	6	28.5	72	1~6	1229.99	1225.90	1223.80	27.57
8	6	28.5	72	1~6	9839.90	9807.8	9790.3	220.57
1	6	78.4	198	1~6	9392.03	9275.90	9261.00	136.99
1	4	2.52	8	1~4	7.80	7.79	7.76	0.91
8	4	2.52	8	1~4	62.41	62.585	62.335	7.276
256	4	2.52	8	1~4	1997.00	2002.7	1994.7	232.83
1	4	4.4	14	1~4	23.99	23.92	23.85	2.01
2	4	4.4	14	1~4	47.97	47.9	47.76	4.0135
8	4	4.4	14	1~4	191.89	191.6	191.04	16.054
64	4	4.4	14	1~4	1535.09	1532.8	1528.3	128.43
256	4	4.4	14	1~4	6140.36	6131.1	6113.2	513.72
1	4	9.5	28	1~4	103.59	103.42	103.19	5.35
8	4	9.5	28	1~4	828.70	827.62	825.79	42.811
64	4	9.5	28	1~4	6629.58	6621	6606.3	342.49
1	4	28.5	72	1~4	800.71	798.08	796.65	23.38
8	4	28.5	72	1~4	6405.64	6384.9	6373.5	187.02
1	4	78.4	198	1~4	6092.66	6038.60	6028.90	125.26
1	2	2.52	8	1~2	3.68	3.6936	3.6788	0.62
8	2	2.52	8	1~2	29.40	29.802	29.683	4.9591
256	2	2.52	8	1~2	940.83	953.67	949.86	158.69
1	2	4.4	14	1~2	11.32	11.373	11.34	1.38
2	2	4.4	14	1~2	22.63	22.809	22.743	2.7523
8	2	4.4	14	1~2	90.53	91.237	90.971	11.009
64	2	4.4	14	1~2	724.21	729.9	727.76	88.073
256	2	4.4	14	1~2	2896.86	2919.6	2911.1	352.29
1	2	9.5	28	1~2	49.28	49.232	49.123	3.87
2	2	9.5	28	1~2	98.55	98.527	98.308	7.7367

8	2	9.5	28	1~2	394.21	394.11	393.23	30.947
64	2	9.5	28	1~2	3153.68	3152.9	3145.9	247.57
1	2	28.5	72	1~2	382.90	380.02	379.34	18.98
8	2	28.5	72	1~2	3063.19	3040.4	3035	151.83
1	2	78.4	198	1~2	2921.25	2875.5	2870.9	112.87

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for non-shielding and metal_ring_flag=0		
					Cmain	As- Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	5	2.52	8	2~6	10.02036523	10.07	9.921	0.63747
1	5	4.4	14	2~6	31.00783926	31.154	30.728	1.3505
1	5	9.5	28	2~6	133.9939609	135.18	133.42	3.5556
1	5	28.5	72	2~6	1035.639188	1045.1	1031.9	15.731
1	5	78.4	198	2~6	7838.32279	7916.1	7817.7	86.064
1	4	2.52	8	3~6	8.099240703	8.0655	7.9703	0.46957
1	4	4.4	14	3~6	24.99391992	24.967	24.699	1.0093
1	4	9.5	28	3~6	108.0861372	108.35	107.26	2.6859
1	4	28.5	72	3~6	837.1765975	837.74	829.67	12.012
1	4	78.4	198	3~6	6397.12679	6345.5	6285.4	66.246
1	2	2.52	8	5~6	4.015562266	3.9768	3.9494	0.27435
1	2	4.4	14	5~6	12.42262789	12.344	12.272	0.57084
1	2	9.5	28	5~6	54.14493469	53.622	53.348	1.5572
1	2	28.5	72	5~6	421.0755925	414.71	412.75	7.6239
1	2	78.4	198	5~6	3219.15638	3141.3	3127	45.756
1	4	78.4	198	2~5	6230.20421	6345.5	6285.4	83.33
1	2	78.4	198	2~3	3029.95338	3141.3	3127	77.037
1	6	2.52	8	1~6	11.94744504	11.985	11.937	0.85684
1	6	4.4	14	1~6	36.81832367	36.758	36.65	1.8104
1	6	9.5	28	1~6	159.0841695	158.88	158.53	4.796
1	6	28.5	72	1~6	1229.987608	1225.9	1223.8	21.593
1	6	78.4	198	1~6	9392.02679	9275.9	9261	120.11
1	4	2.52	8	1~6	7.800766016	7.7914	7.7602	0.66974
1	4	4.4	14	1~6	23.98579211	23.918	23.848	1.4669
1	4	9.5	28	1~6	103.5871266	103.42	103.19	4.0614
1	4	28.5	72	1~6	800.7054375	798.08	796.65	19.521
1	4	78.4	198	1~6	6092.66458	6038.6	6028.9	114.37
1	2	2.52	8	1~6	3.675110781	3.6936	3.6788	0.52495
1	2	4.4	14	1~6	11.31585563	11.373	11.34	1.1257
1	2	9.5	28	1~6	49.27629344	49.232	49.123	3.2291

1	2	28.5	72	1~6	382.8989725	380.02	379.34	17.009
1	2	78.4	198	1~6	2921.25278	2875.5	2870.9	107.26

Table 2-9 Simulation results of Stripe type with shielding layer MOM capacitor Model

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for shielding and metal_ring_flag=1		
					Cmain	As- Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	5	2.52	8	2~6	10.02036523	10.07	9.92	1.3886
256	5	2.52	8	2~6	2565.2135	2586	2547.9	355.49
1	5	4.4	14	2~6	31.00783926	31.15	30.73	3.1581
8	5	4.4	14	2~6	248.0627141	249.49	246.07	25.265
64	5	4.4	14	2~6	1984.501713	1995.9	1968.6	202.12
256	5	4.4	14	2~6	7938.00685	7983.6	7874.4	808.47
1	5	9.5	28	2~6	133.9939609	135.18	133.42	9.1231
64	5	9.5	28	2~6	8575.6135	8653.3	8540.7	583.88
1	5	28.5	72	2~6	1035.639188	1045.10	1031.90	46.309
8	5	28.5	72	2~6	8285.1135	8361.1	8255.7	370.47
1	5	78.4	198	2~6	7838.32279	7916.10	7817.70	281.51
1	4	2.52	8	3~6	8.099240703	8.07	7.97	1.2306
8	4	2.52	8	3~6	64.79392563	64.777	64.015	9.8449
256	4	2.52	8	3~6	2073.40562	2072.9	2048.5	315.04
1	4	4.4	14	3~6	24.99391992	24.97	24.70	2.8459
2	4	4.4	14	3~6	49.98783984	49.997	49.461	5.6918
8	4	4.4	14	3~6	199.9513594	199.99	197.84	22.767
64	4	4.4	14	3~6	1599.610875	1599.9	1582.8	182.14
256	4	4.4	14	3~6	6398.4435	6399.6	6331	728.55
1	4	9.5	28	3~6	108.0861372	108.35	107.26	8.4258
8	4	9.5	28	3~6	864.6890975	867.05	858.34	67.407
64	4	9.5	28	3~6	6917.51278	6936.4	6866.7	539.25
1	4	28.5	72	3~6	837.1765975	837.74	829.67	44.298
8	4	28.5	72	3~6	6697.41278	6702.1	6637.6	354.38
1	4	78.4	198	3~6	6397.12679	6345.50	6285.40	275.91
1	2	2.52	8	5~6	4.015562266	3.98	3.95	0.94225
8	2	2.52	8	5~6	32.12449813	32.068	31.848	7.538
256	2	2.52	8	5~6	1027.98394	1026.2	1019.1	241.21
1	2	4.4	14	5~6	12.42262789	12.34	12.27	2.2008

2	2	4.4	14	5~6	24.84525578	24.751	24.607	4.4017
8	2	4.4	14	5~6	99.38102313	99.003	98.43	17.607
64	2	4.4	14	5~6	795.048185	792.03	787.44	140.85
256	2	4.4	14	5~6	3180.19274	3168.1	3149.8	563.42
1	2	9.5	28	5~6	54.14493469	53.35	53.35	6.8894
2	2	9.5	28	5~6	108.2898694	107.31	106.76	13.779
8	2	9.5	28	5~6	433.1594775	429.23	427.04	55.115
64	2	9.5	28	5~6	3465.27582	3433.9	3416.3	440.92
1	2	28.5	72	5~6	421.0755925	414.71	412.75	39.716
8	2	28.5	72	5~6	3368.60474	3317.9	3302.3	317.73
1	2	78.4	198	5~6	3219.15638	3127.00	3127.00	263.02
256	2	4.4	14	5~6	3032.83854	3168.1	3149.8	547.21
64	2	9.5	28	5~6	3294.04562	3433.9	3416.3	431.95
8	2	28.5	72	5~6	3192.22562	3317.9	3302.3	314.45
1	2	78.4	198	5~6	3029.95338	3141.30	3127.00	261.82

mh	nm	l(um)	nf	Stack	Silicon	SPICE Model for shielding and metal_ring_flag=0		
					Cmain	As- Extracted(fF)	Centering(fF)	Cox (per port)(fF)
1	5	2.52	8	2~6	10.020365	10.07	9.921	1.0883
1	5	4.4	14	2~6	31.007839	31.154	30.728	2.4818
1	5	9.5	28	2~6	133.99396	135.18	133.42	7.507
1	5	28.5	72	2~6	1035.6392	1045.1	1031.9	41.478
1	5	78.4	198	2~6	7838.3228	7916.1	7817.7	267.87
1	4	2.52	8	3~6	8.0992407	8.0655	7.9703	0.98317
1	4	4.4	14	3~6	24.99392	24.967	24.699	2.2888
1	4	9.5	28	3~6	108.08614	108.35	107.26	7.0946
1	4	28.5	72	3~6	837.1766	837.74	829.67	40.318
1	4	78.4	198	3~6	6397.1268	6345.5	6285.4	264.68
1	2	2.52	8	5~6	4.0155623	3.9768	3.9494	0.8295
1	2	4.4	14	5~6	12.422628	12.344	12.272	1.9229
1	2	9.5	28	5~6	54.144935	53.622	53.348	6.1986
1	2	28.5	72	5~6	421.07559	414.71	412.75	37.613
1	2	78.4	198	5~6	3219.1564	3141.3	3127	257.04

1	4	78.4	198	2~6	6230.2042	6345.5	6285.4	264.62
1	2	78.4	198	2~3	3029.9534	3141.3	3127	256.89

3 Appendix

3.1 HSPICE Warning Message

Table 3-1 HSPICE warning message

NO.	Warning Message	Explanation
1.	None	None

3.2 ELDO Warning Message

Table 3-2 Eldo warning message

NO.	Warning Message	Explanation
1.	None	None

3.3 SPECTRE Warning Message

Table 3-3 Spectre warning message

NO.	Warning Message	Explanation
1.	None	None